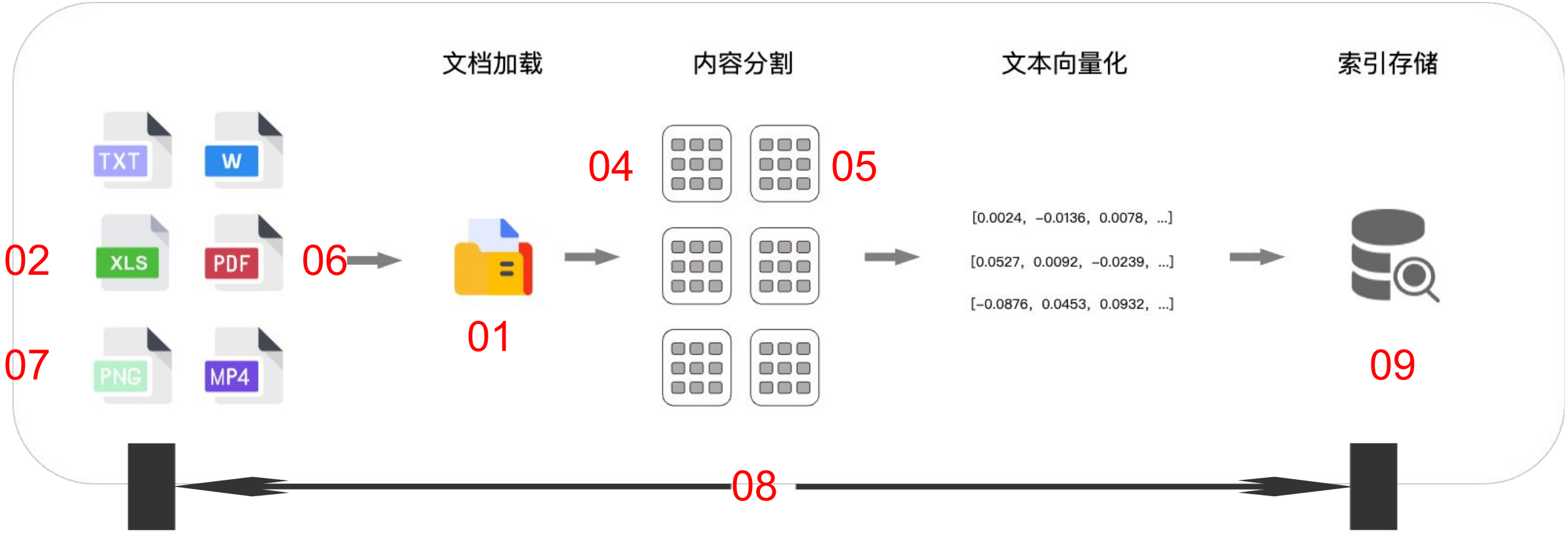
技术点分布与架构汇总

课程涉及的RAG技术

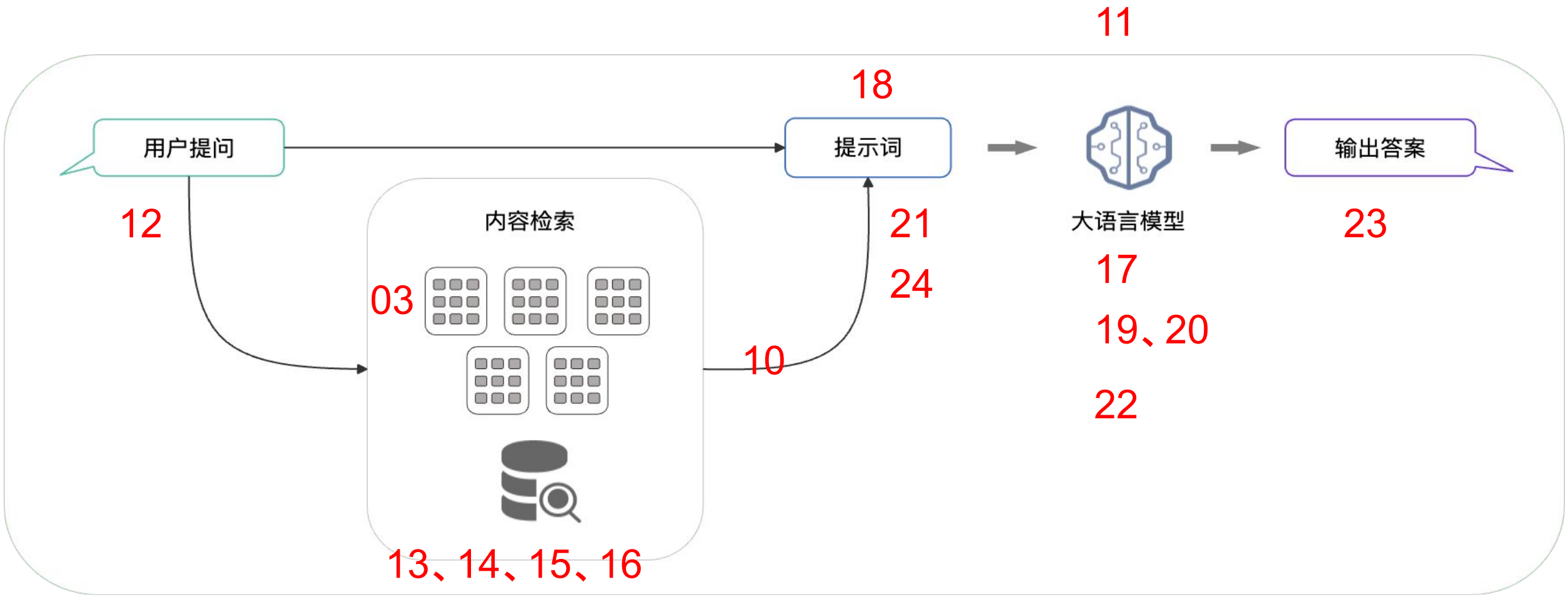
- Naive RAG
- Advanced RAG
- Modular RAG
- Graph RAG
- Multimodal RAG
- CRAG

- llamaindex
- langchain（langgraph、langsmith）
- Dify
- 设计范式（RAG、Agent）
- 系统架构（模型层、服务管理）

索引阶段


文档加载
内容分割
文本向量化
索引存储
TXT
W
XLS
PDF
PNG
MP4
02
06
07
01
04
05
[0.0024, -0.0136, 0.0078, ...]
[0.0527, 0.0092, -0.0239, ...]
[-0.0876, 0.0453, 0.0932, ...]
09
08

检索与生成


11
18
用户提问
提示词
输出答案
12
内容检索
21
大语言模型
23
03
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17
19、20
10
22
13、14、15、16

Agentic RAG

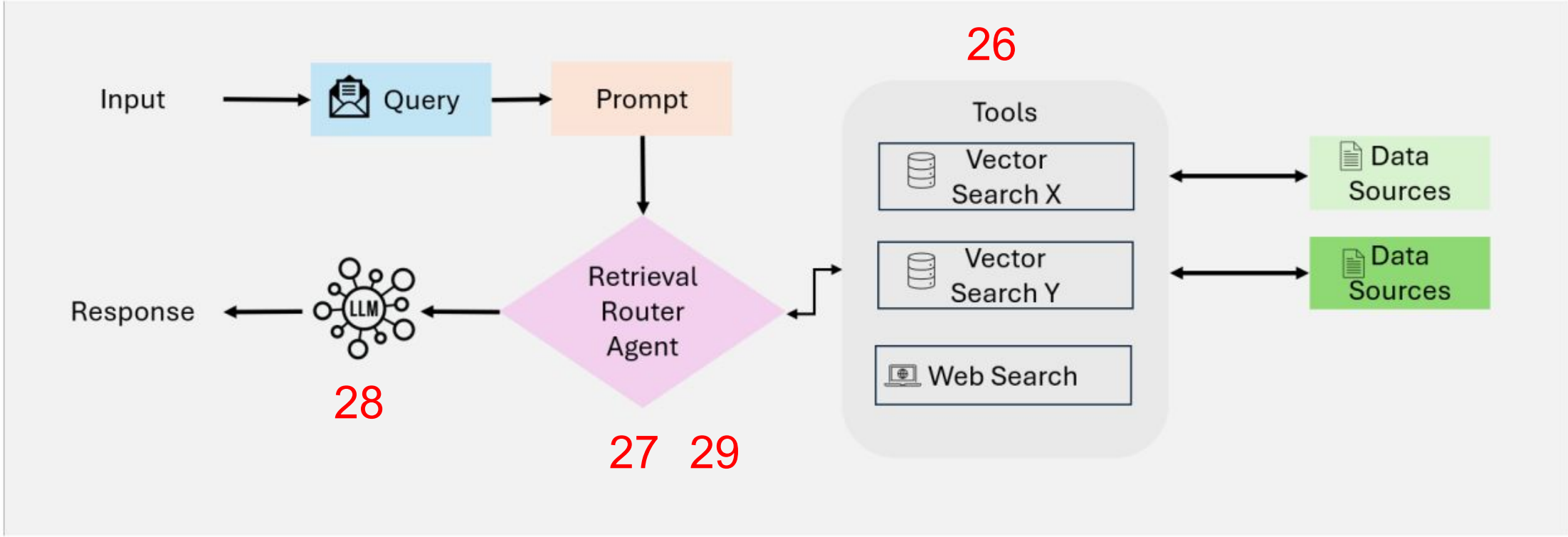

Figure 16: An Overview of Single Agentic RAG

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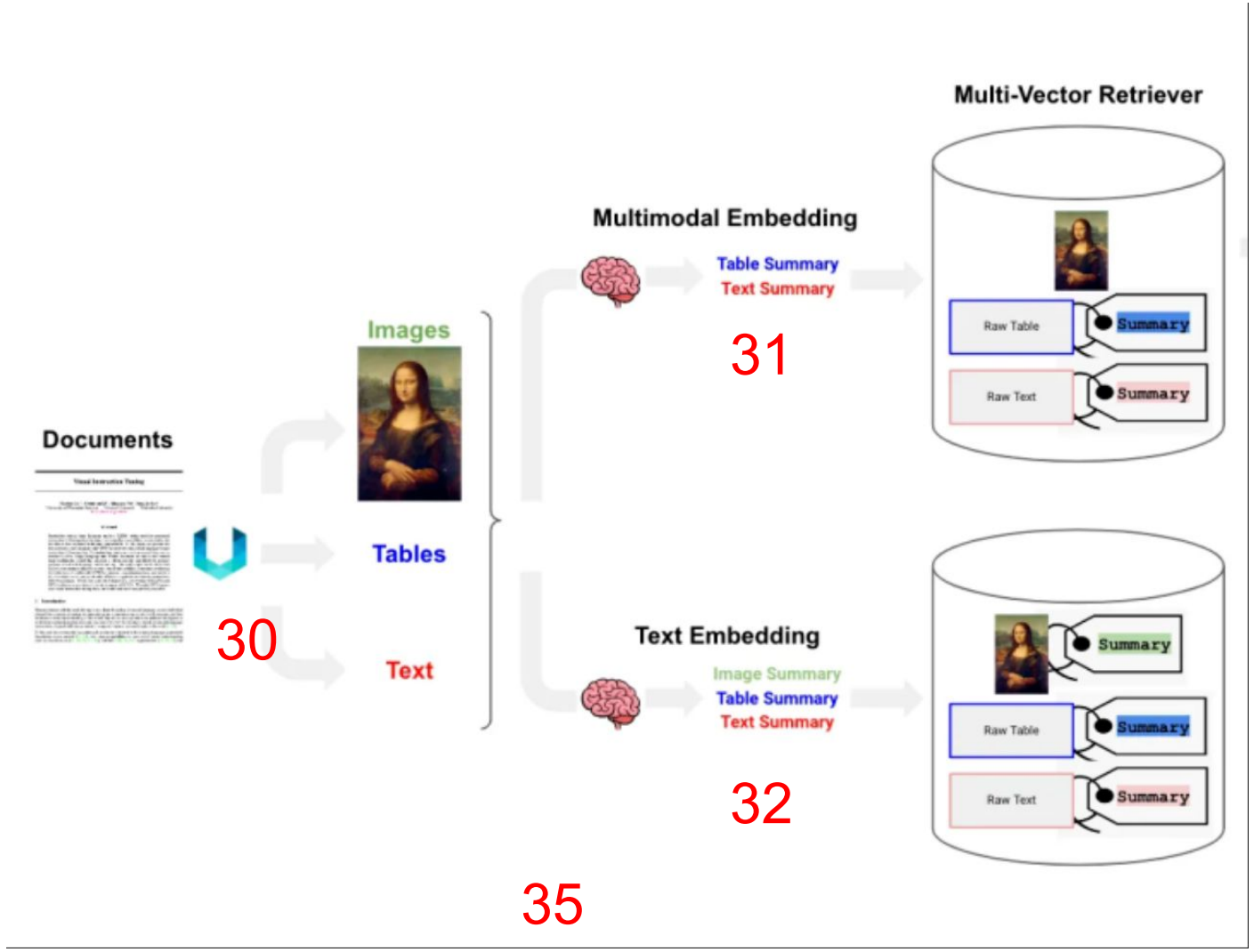
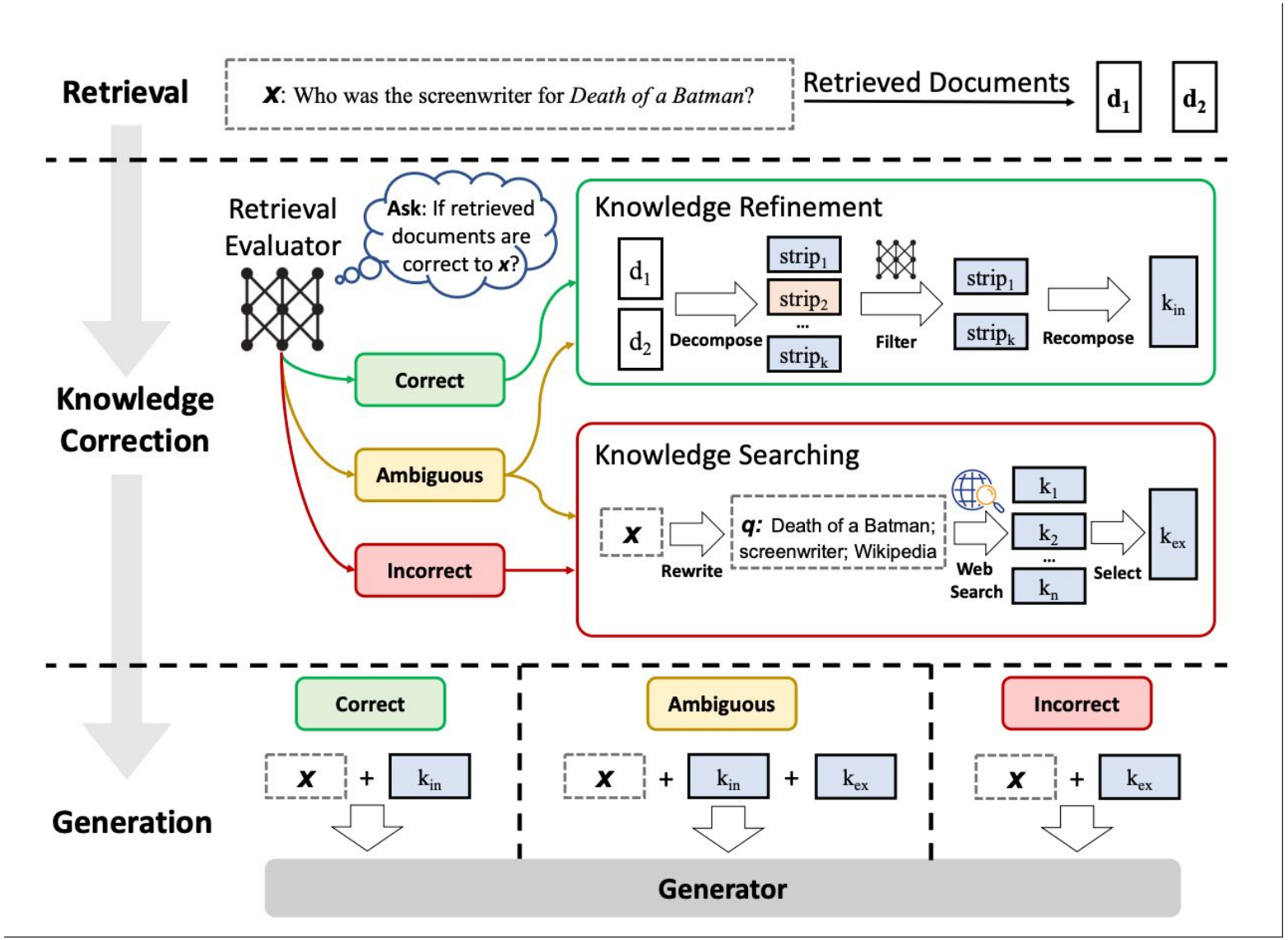
Multimodal RAG



CRAG


Retrieval
x: Who was the screenwriter for Death of a Batman?
Retrieved Documents
d1
d2
Retrieval Evaluator
Ask: If retrieved documents are correct to x?
Knowledge Correction
Correct
Ambiguous
Incorrect
Knowledge Refinement
d1
d2
Decompose
strip1
strip2
...
stripk
Filter
strip1
stripk
Recompose
kin
Knowledge Searching
x
Rewrite
q: Death of a Batman; screenwriter; Wikipedia
Web Search
k1
k2
...
kn
Select
kex
Generation
Correct
x + kin
Ambiguous
x + kin + kex
Incorrect
x + kex
Generator

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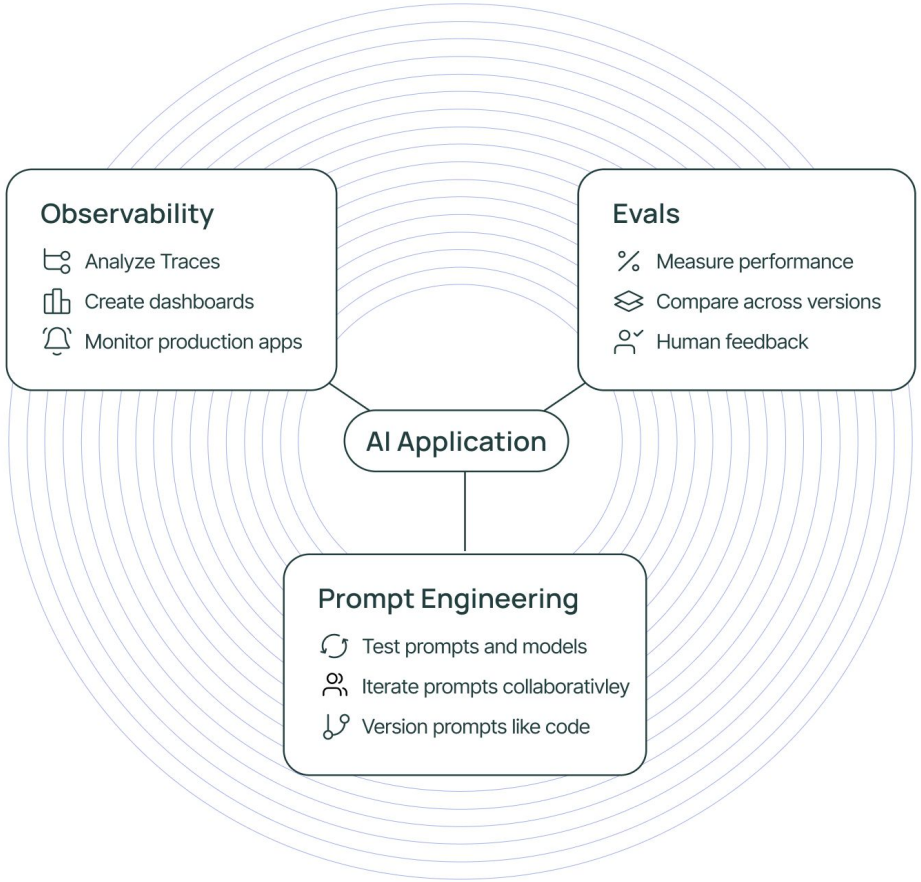
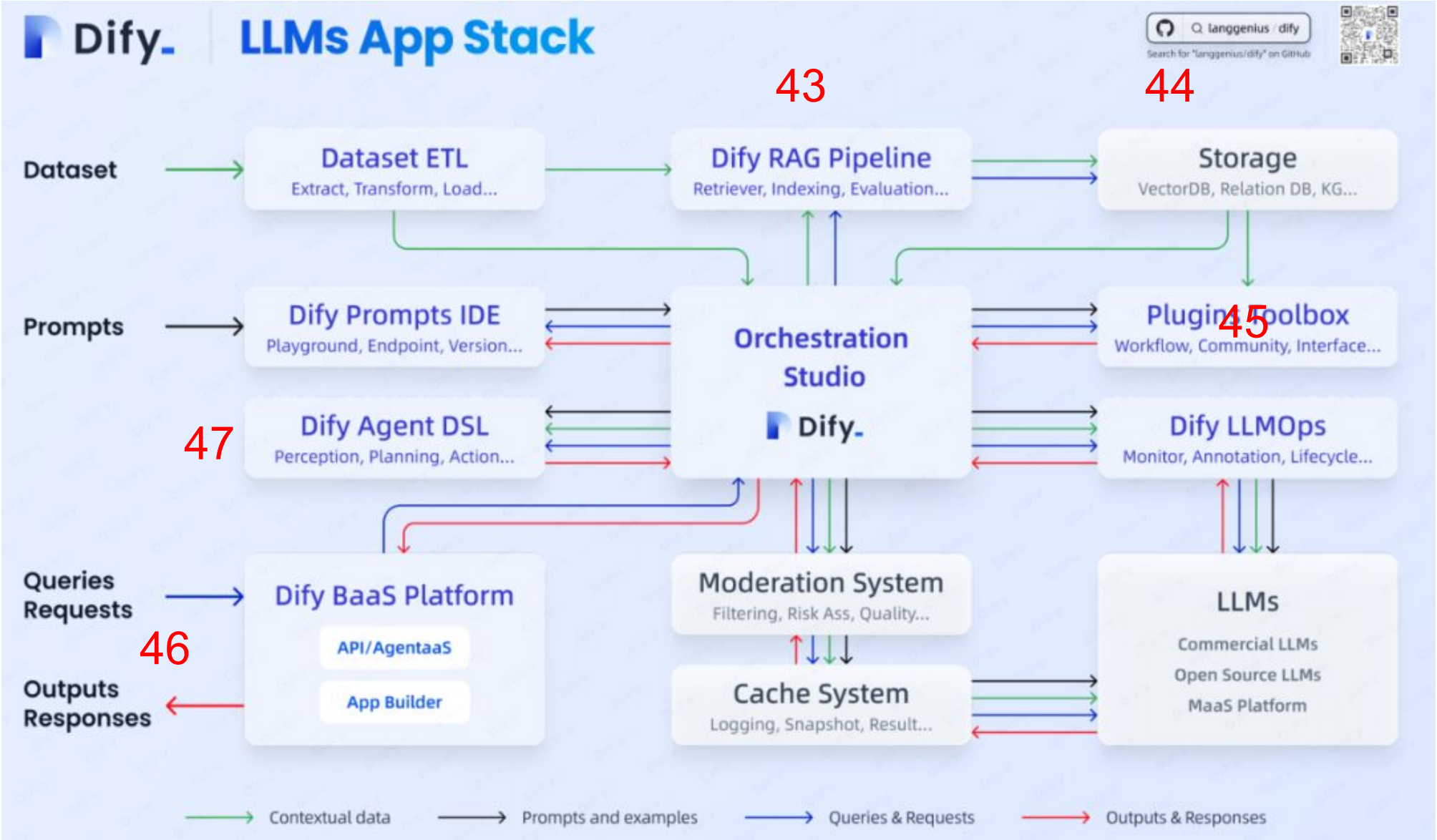
LangSmith

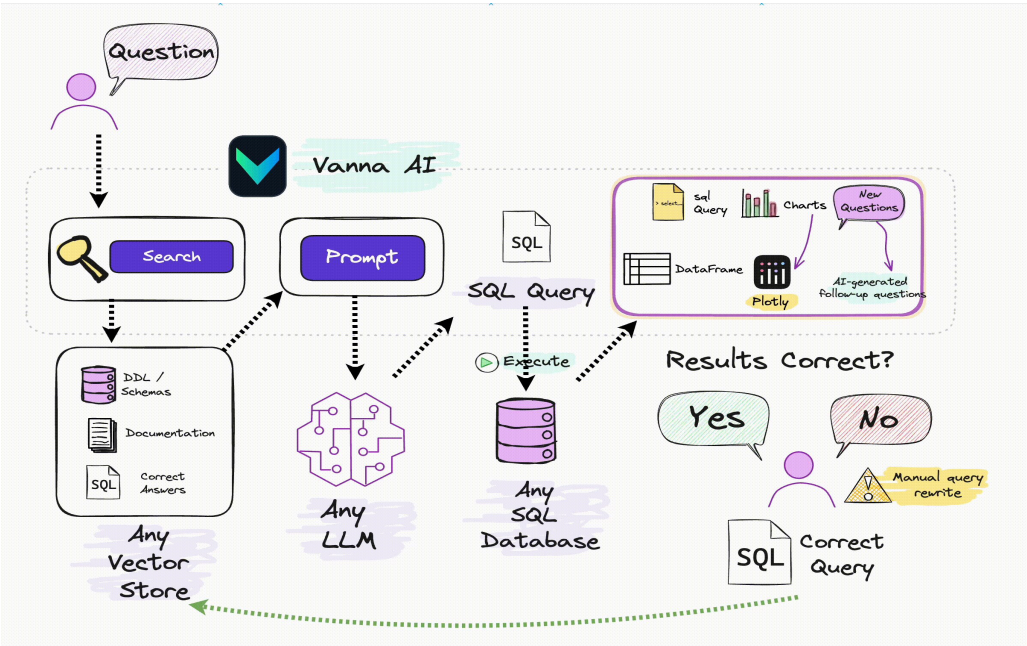
40、41

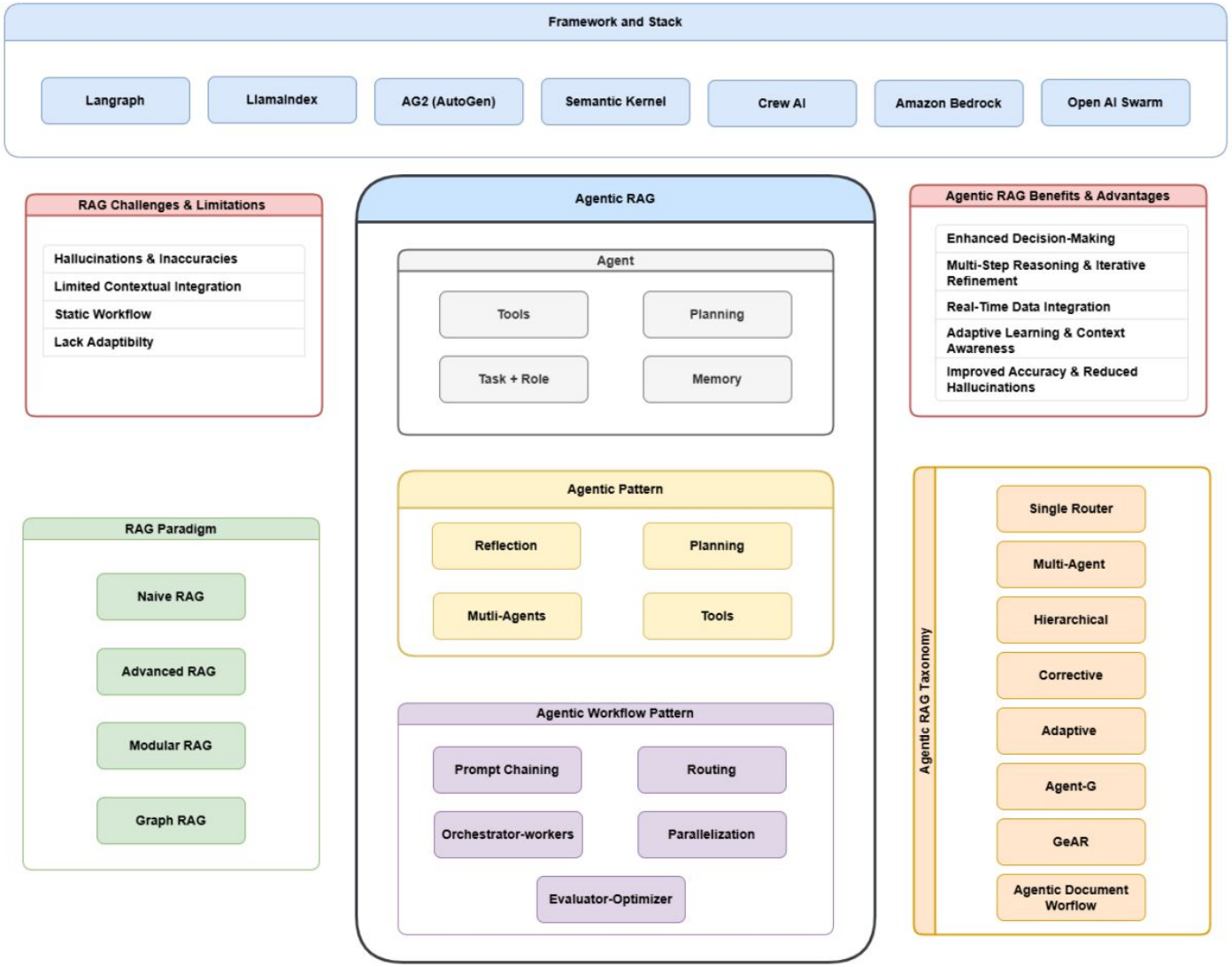


Dify


Dify.
LLMs App Stack
langgenius / dify
Search for "langgenius/dify" on GitHub
43
44
Dataset
Dataset ETL
Extract, Transform, Load...
Dify RAG Pipeline
Retriever, Indexing, Evaluation...
Storage
VectorDB, Relation DB, KG...
Prompts
Dify Prompts IDE
Playground, Endpoint, Version...
Orchestration Studio
Dify.
Plugins Toolbox
Workflow, Community, Interface...
45
42
47
Dify Agent DSL
Perception, Planning, Action...
Dify LLMOps
Monitor, Annotation, Lifecycle...
Queries Requests
Dify BaaS Platform
API/AgentaaS
App Builder
46
Moderation System
Filtering, Risk Ass, Quality...
Cache System
Logging, Snapshot, Result...
LLMs
Commercial LLMs
Open Source LLMs
MaaS Platform
Outputs Responses
Contextual data
Prompts and examples
Queries & Requests
Outputs & Responses

设计范式


Question
Vanna AI
Search
Prompt
SQL
SQL Query
sql Query
Charts
New Questions
DataFrame
Plotly
AI-generated follow-up questions
Execute
Results Correct?
DDL / Schemas
Documentation
Correct Answers
Any Vector Store
Any LLM
Any SQL Database
Yes
No
Manual query rewrite
SQL
Correct Query


Framework and Stack
Langraph
Llamaindex
AG2 (AutoGen)
Semantic Kernel
Crew AI
Amazon Bedrock
Open AI Swarm
RAG Challenges & Limitations
Hallucinations & Inaccuracies
Limited Contextual Integration
Static Workflow
Lack Adaptibilty
RAG Paradigm
Naive RAG
Advanced RAG
Modular RAG
Graph RAG
Agentic RAG
Agent
Tools
Planning
Task + Role
Memory
Agentic Pattern
Reflection
Planning
Mutli-Agents
Tools
Agentic Workflow Pattern
Prompt Chaining
Routing
Orchestrator-workers
Parallelization
Evaluator-Optimizer
Agentic RAG Benefits & Advantages
Enhanced Decision-Making
Multi-Step Reasoning & Iterative Refinement
Real-Time Data Integration
Adaptive Learning & Context Awareness
Improved Accuracy & Reduced Hallucinations
Agentic RAG Taxonomy
Single Router
Multi-Agent
Hierarchical
Corrective
Adaptive
Agent-G
GeAR
Agentic Document Worflow

架构分层

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38、39